P08

Tellegen's Theorem and Power Invariance in Lumped Networks

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Domain DOM_ENG

Abstract

For any lumped network obeying Kirchhoff's laws, the inner product of branch voltages and currents is zero: sum(v_k * i_k) = 0. The proof relies solely on orthogonal subspace projection between cutsets and cycles.

Abstract quoted from B. D. H. Tellegen et al., Philips Research Reports (1952).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?